Orphans and Archetypes: Placing Unassigned Hittite Ritual Fragments in Their Traditions

Anonymous ACL submission

Abstract

Thousands of Hittite ritual and festival fragments cannot be assigned to a composition because no parallel passage survives. Laroche's residual category CTH 470 and the recently digitised EBo fragments hold 2,944 such manuscripts in the TLHdig 0.3 release, with a median of 9 legible words in CTH 470 and 3 among the EBo. We ask whether the words of a fragment and the way they are spelled can place it in a ritual or festival tradition, and how far such a placement can be trusted. Ten reference traditions are taken from the studies that argued for them and mapped to Laroche's CTH numbers. Nearest-centroid assignment over tf-idf features of word forms and sign sequences returns 82% of held-out manuscripts to their tradition (majority baseline 29%), and 56 of 78 compositions held out whole return to their scholarly group. A margin calibrated on held-out fragments of matching size lets the method abstain; it places 380 of 1,766 orphans at an expected precision of 80%. Splitting each decision exactly over its features shows that traditions are identified by what is written, above all agents, deities, realia and foreign-language words, and hardly by how it is spelled; common particles make up 47% of the similarity but 11% of the decision. Code and outputs are released, and every number in the paper is generated from them.

1 Introduction

Emmanuel Laroche's *Catalogue des textes hittites* (Laroche, 1971) gave every Hittite composition a number, and with it a place: rituals in one block, festivals in another, texts in Hattian, Luwian and Hurrian in blocks of their own. What could not be placed went into residual categories. CTH 470, "fragments of rituals", is the largest of them, and it has grown with every volume of hand copies since 1971. The Mainz Hethitologie-Portal keeps Laroche's numbers as a shared address system while moving fragments after joins (Košak et al., 2026; Košak, 2026). The digitisation of the EBo fragments in TLHdig 0.3 (Müller et al., 2026) adds 1,110 more manuscripts with a median of 3 legible words, most of them filed under two further residual numbers, CTH 832 and CTH 670.

A philologist assigns such a fragment through a parallel: a duplicate, or a close parallel passage in an identified composition. For most of these orphans that route is closed by definition. We ask instead whether the content of a fragment, its words and their spellings, can point to its tradition, and how far such a pointer can be trusted. Put this way, the task is attribution, and two known weaknesses of attribution apply with full force: accuracy falls with the length of the sample (Eder, 2015), and a classifier with a closed set of candidates always names one of them, even when the true source is not among them (Koppel et al., 2011). A placement is also only useful to a philologist who can check it, which requires knowing what it rests on.

Gordin et al. (2025) mapped about 1,000 Hurrian and Luwian festival manuscripts with tf-idf and t-SNE and proposed, as a next step, to test which word groups carry the signal. We turn that exploration into assignment, with explicit baselines, held-out evaluation and a rule for when not to assign. Our contributions are:

1. a reference of ten ritual and festival traditions, each defined by the study that argued for it and mapped to Laroche's CTH numbers, with the ranges that no study claims set aside as test sets (Section 3);

2. an evaluation that holds out manuscripts and whole compositions, so that the scholarly groupings themselves, and one contested attribution, are tested; that follows accuracy down to fragment size; and that calibrates a margin for abstention (Section 5);

3. an exact decomposition of each decision over word categories and writing layers, which separates what is written from how it is written and shows how much rests on common particles such as *nu* (Section 5.4);

4. a placement of 380 orphans, each with the features it rests on, as leads for verification on the tablet.

2 Background

Traditions and the catalogue. Laroche sorted by genre first and by language second, while later scholarship defines a ritual tradition by the origin of its practitioners, its cult and its ritual language. One tradition therefore ends up scattered over several CTH ranges. The Kizzuwatna rituals occupy CTH 471–500, but the ritual of the Kizzuwatnean Old Woman Maštigga is filed with the named practitioners at CTH 404 (Miller, 2004); the rituals of the AZU priest (CTH 701) sit in the Hurrian block (Salvini and Wegner, 1986); and the named-practitioner rituals of CTH 390–469 come from Arzawa, the Lower Land and Kizzuwatna alike (Christiansen, 2006; Bawanypeck, 2005). The festivals have been regrouped by cult geography (Gordin et al., 2025; Görke, 2022; Trameri, 2025) and, in the Hethitische Festrituale project (HFR), into edition modules such as the Luwian and the Palaic festivals (Rieken and Schwemer, 2026; Görke, 2020).

Scribal habit and attribution. Hittite scribes copied, excerpted and re-edited ritual texts over two centuries, and their habits of logographic writing, plene spelling and sign choice vary with date, school and individual (Gordin, 2015; Weeden, 2011; van den Hout, 2009; Waal, 2015). Whether a habit marks a tradition, a circle of scribes or a date is therefore a question to test, not an assumption. Stylometry supplies frequency-based representations, distance measures and protocols with held-out evaluation (Burrows, 2002; Juola, 2008; Stamatatos, 2009), and open-set recognition the means to reject a sample that fits no candidate (Scheirer et al., 2013).

Computational work on cuneiform. Machine learning on cuneiform has addressed transliteration and translation (Gordin et al., 2020; Yavasan and Gordin, 2025), language and dialect identification (Jauhiainen et al., 2019), the matching of fragments to known compositions by sign sequences (Simonjetz et al., 2024), and palaeographic style (Yugay et al., 2024). Fragment matching finds a known text; it does not place a fragment whose parallels are lost, and all of these studies are closed-set. We know of no published study that uses content features to assign Hittite fragments to religious traditions, or that asks when a fragment fits none of them.

3 Data

Corpus. We use the TLHdig Beta 0.3 release (Müller et al., 2026), converted to one row per word following Yavasan and Gordin (2025). Each word carries its normalised form, its transliteration split into signs, its language, whether a sign is an Akkadogram, a Sumerogram or a determinative (TLHdig tags these separately), and the lexical and morphological analyses that TLHdig produces automatically with the HFR tool, which are not validated by hand (Müller et al., 2026). The release is downloaded by script and checked by its checksum, so the word table can be rebuilt from the public data alone. As in Gordin et al. (2025), reference manuscripts must have at least 17 legible words. The analyses do not cover the whole release: the Luwian rituals (CTH 758–763) have none, and many other rituals lack them. The main model reads analysed words, so in its reference the Luwian rituals have 7 manuscripts and the western rituals 8; Section 5.3 repeats the evaluation on spellings alone with every manuscript.

Damage and restoration. TLHdig prints the editors' restorations alongside what survives, and marks each sign as preserved, damaged, restored, omitted or lost. Counting restorations as text would let an editor's reconstruction, often made from a duplicate, decide where a fragment belongs. We therefore count only what is on the tablet. A sign counts if it is preserved or damaged, also when its reading is uncertain; a word counts if at least half its signs survive, or if the same word form survives at least half-preserved elsewhere in the same text. Restorations supported only by duplicates are not used. Sign n-grams never span a missing sign, and a spelling counts only when every sign of the word survives. We report the editors' full text, restorations included, as a comparison (*version A*); the main results use the damage-aware text (*version C*).

Orphans. Orphans are all CTH 470 manuscripts and all EBo manuscripts, wherever TLHdig files them, 2,944 in all. CTH 670 and 832, where most EBo fragments are filed, are pools of orphans, not

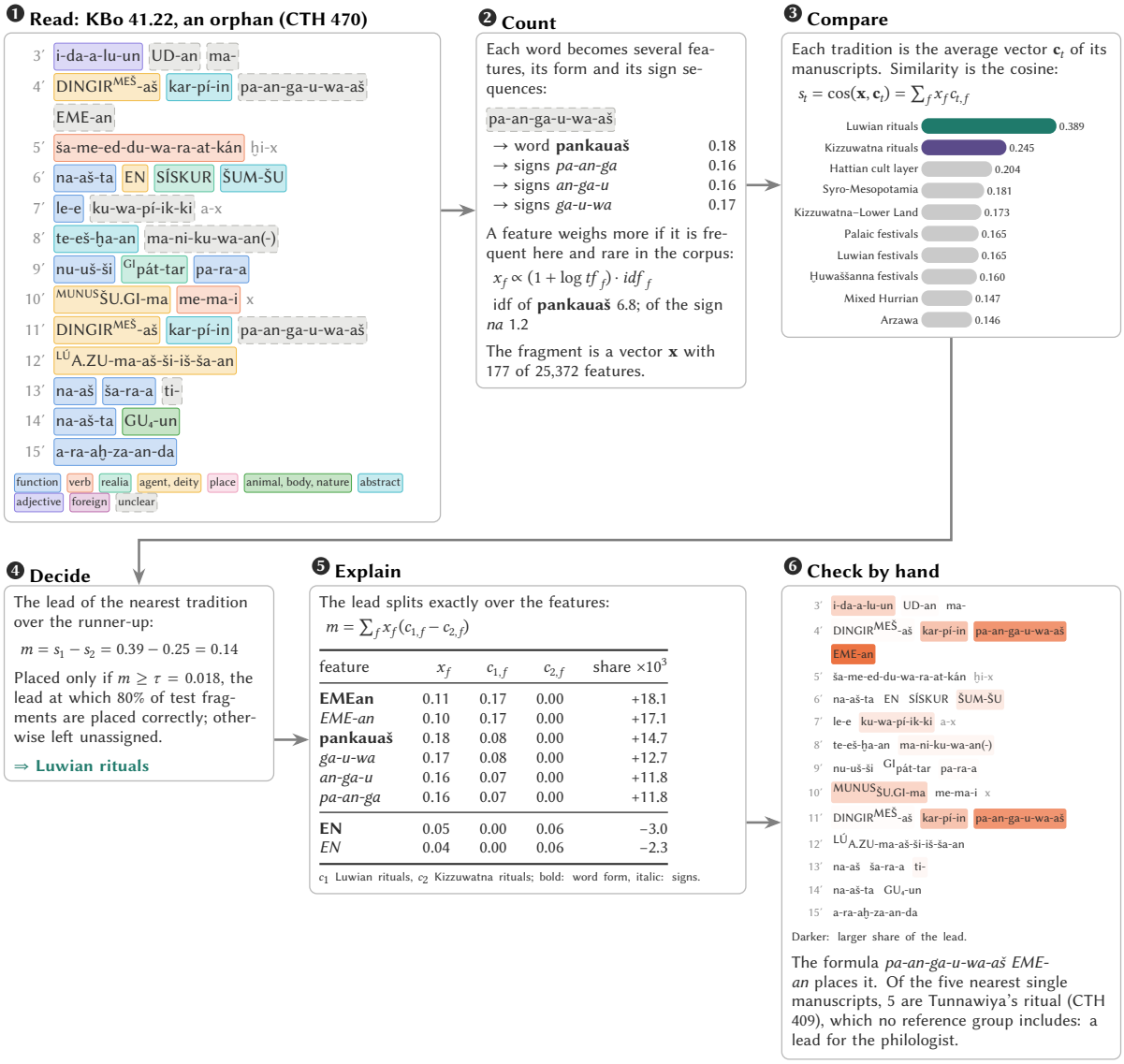


Figure 1: The worked example in version A (the editor's full text): the orphan KBo 41.22 (filed under CTH 470 in TLHdig 0.3) from its text to its placement. In the damage-aware version C it is not placed (Section 4). (1) The legible words, tinted by word category (dashed: a low-confidence or unclear class). (2) Each word becomes a word-form feature and sign-sequence features, weighted by tf-idf. (3) Cosine similarity to the centroid of each tradition. (4) The lead of the nearest tradition over the runner-up, against the calibrated margin. (5) The lead split over the features that make it. (6) The same split on the text, and the closest single manuscripts as leads for the philologist. All values are generated from the experiment outputs.

traditions. Of the 2,814 orphans in the CTH ranges we load, 1,766 have at least five legible words and are placed (median 6 words). Orphans never enter the reference.

Reference traditions. Each tradition is defined by the study that argued for it (Table 1). The Hurrian festival groups follow Table 3 of Gordin et al. (2025). Its single Luwian group is split as HFR splits it, into the festivals of Ištanuwa and Lallupiya and those of Ḫuwaššanna, and HFR's Palaic festivals are added (Rieken and Schwemer, 2026). The ritual traditions are the Kizzuwatna rituals as delimited by Miller (2004); the western rituals, that is, the augur rituals (Bawanypeck, 2005) and those whose colophon names Arzawa or Ḫapalla as the practitioner's home; the Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi (CTH 758–763), which Yakubovich and Mouton (2023) place in the Lower Land rather than in Kizzuwatna; and the Hattian cult layer of Klinger (1996), whose texts we take from the citations of his study in the Mainz Konkordanz (Košak, 2026). The Ambazzi rituals (CTH 391,

Tradition	CTH	Defining studies	n
Kizzuwatna and Lower Land (Hurr.)	628, 699–700, 711–713, 719, 722	Gordin et al. (2025) Table 3; Wegner and Salvini (1991)	276
Syro-Mesopotamia (Hurr.)	698, 701, 714–715, 721	Gordin et al. (2025) Table 3; Salvini and Wegner (1986)	134
Mixed (Hurr., Syr., Luw.)	704–706, 720	Gordin et al. (2025) Table 3	170
Ištanuwa and Lallupiya festivals	665, 771–773	Rieken and Schwemer (2026); Görke (2020); Hutter (2003)	47
Ḫuwaššanna festivals	690–694	Rieken and Schwemer (2026); Hutter (2003)	152
Palaic festivals	750–754	Rieken and Schwemer (2026); Görke (2020)	30
Kizzuwatna rituals	404, 471–499	Miller (2004); Strauß (2006); Görke (2010)	140
Arzawa (western) rituals	394, 402, 406, 410, 425, 433	Bawanypeck (2005); colophons	8
Luwian rituals (Lower Land)	758–763	Yakubovich and Mouton (2023)	7
Hattian cult layer	591, 627, 725–729, 731–746, 820–821	Klinger (1996)	386

Table 1: Reference traditions, the studies that define them, and the number of reference manuscripts with 17 or more legible words. CTH numbers follow the Portal CTH (Košak et al., 2026).

429, 463) show Luwian influence but no stated link to Arzawa (Christiansen, 2006), so they stay outside the western group; Section 5.2 tests that decision.

Two sets leave the reference and are placed like orphans: the texts of CTH 390–469 that no defining study claims, and CTH 500, "fragments of festival and incantation rituals from Kizzuwatna", which is a running list of fragments rather than a composition. For CTH 701, the subgroups a–f of Salvini and Wegner (1986) and the script dates come from the Konkordanz and serve as partial ground truth.

4 Method

Figure 1 follows one orphan, KBo 41.22, through the procedure with the actual values at each step.

Features. Every manuscript is represented twice (Salton and Buckley, 1988). The word space holds tf-idf weights of normalised word forms and stands for *what* is written; the sign space holds tf-idf weights of sign n-grams (n = 1–3) within words and stands for *how* it is written. The main model concatenates both. Weights are sublinear, rows are L2-normalised, and features found in fewer than two manuscripts are dropped. In version A, KBo 41.22 has 33 legible words, which yield 177 nonzero features out of 25,372.

Assignment and abstention. Each tradition is represented by the normalised centroid of its manuscripts, and a fragment is scored by its cosine similarity to every centroid. It is placed in the most similar tradition when its lead over the runner-up, the *margin*, reaches a threshold τ, and is left unplaced otherwise. We choose τ on held-out data rather than by hand: manuscripts with at least 40 words (at most 40 per tradition) are held out in turn and cut into ten random windows of $k \in \{3, 5, 8, 10, 17, 25, 40\}$ words, and τ is the smallest margin at which these windows reach 80% precision. This gives $\tau = 0.028$ (90% bootstrap interval 0.027–0.030). In version A, where $\tau = 0.018$, KBo 41.22 is closest to the Luwian rituals (0.39) and next to the Kizzuwatna rituals (0.25); its margin of 0.14 is well above τ, and so it is placed. That placement does not survive the damage-aware text. Its decisive words, a formula with DINGIR.MEŠ and *karpin*, are the editor's restorations; only 13 words remain on the tablet, its similarity to the Luwian rituals falls to 0.14, and its lead of 0.013 (Kizzuwatna rituals over Syro-Mesopotamia (Hurrian)) is below $\tau = 0.028$: it is not placed. We keep it as a cautionary example, because it shows each step clearly and how much a placement can owe to restorations. *[A second worked example that version C places is being chosen.]*

Decomposing a decision. Because the score is a dot product, the margin splits exactly into one term per feature, $x_f(c_{\text{top},f} - c_{2,f})$, where x is the fragment and c the two centroids. Summing these terms by word category or layer shows what a placement rests on, and listing the largest ones gives the words a philologist can check (Figure 1, panels 5–6). This matters because similarity and decision differ: common words such as *nu* make up much of a fragment's similarity to every tradition but little of the difference between two.

Validation. Validation is leave-one-manuscript-out: each reference manuscript is removed from its own centroid (in closed form) and reassigned. We report accuracy and per-tradition recall against two baselines, always guessing the largest tradition (ma-

jority) and the same procedure with shuffled labels (200 permutations, which also give a p-value). Idf weights are fitted on the whole reference set.

Layers and word categories. To separate cultural content from scribal habit, the word space is split into layers, each evaluated alone: logograms, Hittite vocabulary, foreign-language words (Hurrian, Luwian, Hattian, Palaic), and the 203 words attested in more than one spelling, once with their spelling kept and once collapsed to the normalised form. If spelling carried tradition beyond the word itself, the first would beat the second. We also use the word categories of Gordin et al. (2025, Table 4): agents, deities, function words, foreign words, realia and verbs, with realia narrowed to physical things and the remaining nouns grouped by meaning (places; animals, body parts and nature; abstract nouns; adjectives; measures).

No word was classed by hand. Fixed rules derive each category from TLHdig's markup, applied in order: the word's determinative (DINGIR makes a deity, ^{LÚ} an agent), its language tag, and its automatic morphological analysis, using the analysis marked as selected where there is one (a person-marked form is a verb, a particle or pronoun a function word); unanalysed words fall back on the analyses selected for the same form elsewhere. Nouns and adjectives are then classed by their German gloss, through a list of 5,494 glosses that was drafted with a language model and has not been checked by hand; it is the one step not derived from TLHdig. Each label carries the confidence of the rule that set it: 63% of the 320,638 legible words are classed with high confidence, 25% with medium and 12% with low. Since neither the automatic analyses nor our rules are validated in TLHdig, our audit is the only manual check: a domain expert checked a stratified sample of 178 words: 93%, 88% and 70% were correct at the three levels, 89% overall when weighted by corpus shares. Fixing the 5 rules that the errors pointed to raises the overall figure to 94%, an upper bound since it is measured on the same sample. The audit scored all nominal words as one class, so it does not cover the split by meaning.

5 Results

5.1 Validation and fragment size

Held out one at a time, 82% of the 1,350 reference manuscripts return to their tradition, against 29% for the majority baseline and 17% for shuffled labels ($p = 0.005$). Words and signs alone do almost as well (79% and 82%). On the editor's full text (version A) the same test gives 84% on 1,626 manuscripts: restorations add manuscripts and words but change little. Recall ranges from 38% to 95% (Table 2) and is lowest for the two smallest traditions. The festivals that HFR separates are kept apart by the texts themselves: the Ḫuwaššanna festivals reach 91% and those of Ištanuwa and Lallupiya 81%.

Tradition	n	recall	comp.	orph.	decided by
Kizzuwatna and Lower Land (Hurr.)	276	78%	4/8	35	realia 36%
Syro-Mesopotamia (Hurr.)	134	66%	0/5	35	realia 37%
Mixed (Hurr., Syr., Luw.)	170	65%	3/4	13	agents, deities 45%
Ištanuwa and Lallupiya festivals	47	81%	2/4	5	agents, deities 40%
Ḫuwaššanna festivals	152	91%	5/5	8	agents, deities 76%
Palaic festivals	30	77%	0/5	1	agents, deities 70%
Kizzuwatna rituals	140	88%	17/18	210	realia 30%
Arzawa (western) rituals	8	38%	0/2	8	–
Luwian rituals (Lower Land)	7	43%	0/2	7	–
Hattian cult layer	386	95%	25/25	58	agents, deities 44%

Table 2: Per tradition: reference manuscripts (n), leave-one-manuscript-out recall, compositions that return to the tradition when held out whole (comp., Section 5.2), orphans placed (orph., Section 5.5), and the word category that carries the largest share of the decision (Section 5.4; not shown below ten correctly placed manuscripts).

Accuracy rises steeply with fragment size and then flattens. Windows of 3, 10 and 40 words cut from held-out manuscripts are placed correctly in 46%, 60% and 73% of cases, against a chance level of 10%. Above the margin $\tau = 0.028$, 49% of the windows are placed, at 80% precision. In version A, where every restored word counts, the same precision is reached at $\tau = 0.018$ and 66% of windows pass.

Inside CTH 701, the procedure recovers 51% of the 72 manuscripts that Salvini and Wegner (1986) sorted into 8 subgroups (majority 19%, $p = 0.005$). The same features do not predict script date (77% against a majority of 81%), so the subgroups are not clusters of date.

5.2 Testing scholarly groupings

Leave-one-manuscript-out accuracy is flattered by duplicates: when one copy of a composition is held out, its other copies stay in the reference. A stricter test removes a whole composition (all manuscripts of one CTH number), rebuilds the centroids, and asks whether the composition returns to the group that scholarship assigned it to. 56 of 78 compositions do, and 783 of 1,351 manuscripts (58%) are placed correctly when their composition is unseen (Table 2). The Hattian cult layer, the Ḫuwaššanna festivals and the Kizzuwatna rituals are recovered as groups. The Hurrian festival groups (7 of 17 together), the Palaic festivals and the two small ritual groups are not: their manuscripts resemble each other because they share compositions, not because their compositions share a profile that sets them apart. Held out whole, the CTH 701 manuscripts go to the Kizzuwatna rituals (0 of 99 to their own group): in content, the rituals of the AZU priest are Kizzuwatna rituals. We read the groups that are not recovered as claims that these features do not support, not as claims that the evidence refutes.

A contested attribution. The Ambazzi rituals (CTH 391, 429, 463) have been read as western because of their Luwian elements (Christiansen, 2006). We pool their 6 manuscripts and score them against every tradition, comparing each tradition's lead over the best other with the same statistic for random sets of the same size drawn from the 264 unclaimed ritual manuscripts (CTH 390–469). The western tradition ranks 6 of 10 ($p = 0.13$). The nearest tradition is the Kizzuwatna rituals, for 5 of the 6 manuscripts, but no nearer than random unclaimed rituals ($p = 0.54$). Of the 20 words most distinctive of the western rituals (Monroe et al., 2008), 1 occurs in the Ambazzi texts (*Éri*). On this evidence the Ambazzi rituals sit in the generic ritual residue, and we keep them out of the western reference.

5.3 Spellings only, with every manuscript

The gap in the annotation falls on the smallest traditions. We therefore repeat the leave-one-out evaluation in the sign space alone, where every legible word counts, once with the analysed manuscripts only and once with all manuscripts of 17 or more legible words (Table 3). The reference grows from 1,349 to 1,902 manuscripts, and overall accuracy falls from 82% to 78%, because most traditions gain manuscripts that are harder to place. The two small ritual traditions gain most: the Luwian rituals grow from 7 to 106 manuscripts and their accuracy from 29% to 87%; the western rituals from 8 to 38 (38% to 74%). Of the 380 orphans the main model places, 216 are also placed with the full reference and 211 of these keep their tradition; 25 move to the Luwian rituals and 5 to the western rituals. Placements in the two small ritual traditions should be read with this in mind.

	analysed only		all manuscripts	
Tradition	n	acc.	n	acc.
Arzawa (western) rituals	8	38%	38	74%
Hattian cult layer	385	95%	527	90%
Kizzuwatna rituals	140	88%	236	81%
Kizzuwatna and Lower Land (Hurr.)	276	78%	323	72%
Ištanuwa and Lallupiya festivals	47	83%	55	74%
Luwian rituals (Lower Land)	7	29%	106	87%
Mixed (Hurr., Syr., Luw.)	170	65%	206	63%
Palaic festivals	30	77%	36	72%
Syro-Mesopotamia (Hurr.)	134	64%	175	63%
Ḫuwaššanna festivals	152	89%	200	84%
All	1,349	82%	1,902	78%

Table 3: Spellings only: leave-one-out accuracy per tradition in the sign space, with the reference limited to manuscripts that have word analyses, and with all manuscripts of 17 or more legible words.

5.4 What a placement rests on

What is written, or how. Each layer alone identifies a tradition (Table 4). Logograms reach 67%, Hittite vocabulary 66%, and foreign-language words 81% in the 546 manuscripts that have them. The variable words reach 45% with their spelling kept and 50% with it collapsed: the choice of spelling adds nothing beyond the word. The seven orthographic rates are a weak but real signal (38% against 28% for shuffled labels). By word category, removing the function words leaves accuracy almost unchanged (78%, against 79% with all words), and on their own they reach 47%, below agents (62%) and foreign words (82%). Removing the foreign words costs most (76%). The formulaic layer is common ground; the traditions differ in whom and what they name.

Similarity is not decision. Splitting each decision over its features (Section 4) answers an obvious worry: the particle *nu* and other frequent words

Layer	n	maj.	shuf.	acc.
Logograms (Sumero- and Akkadograms)	1,295	26%	13%	**67%**
Hittite vocabulary	1,292	26%	15%	**66%**
Foreign-language words (Hurrian, Luwian, Hattian, Palaic)	546	33%	18%	**81%**
Variable words, spelling collapsed	1,215	27%	11%	**50%**
Variable words, spelling kept	1,155	27%	12%	**45%**
Seven orthographic rates (LDA, 10-fold)	1,351	29%	28%	**38%**

Table 4: Each layer alone: leave-one-manuscript-out accuracy, with the majority (maj.) and shuffled-label (shuf.) baselines. n: manuscripts with at least one feature of the layer.

weigh heavily in a fragment's similarity to its tradition. Over the 1,106 correctly placed reference manuscripts, features found in at least a quarter of the reference make up 47% of the similarity but only 11% of the decision, because every centroid shares them and they largely cancel. Deleting the 100 most frequent words, 46% of all word tokens, leaves accuracy at 81%. The decision rests on agents and deities (42%), realia (17%), foreign words (13%) and verbs (12%); function words carry 9%. Each tradition has its own decisive category (Table 2): realia for the Kizzuwatna rituals, agents and deities for the festivals and the Hattian cult layer.

For short fragments the worry is justified. The share of the decision carried by common features falls from 29% for orphans of 5–10 words to 14% above 50 words, and 73 of the 380 placed orphans rest mostly on them. In 10-word windows from reference manuscripts, such placements are right 64% of the time, against 82% for the rest. Without function words, 300 of the placed orphans keep their tradition, 0 move and 80 fall below the margin. Common words thus rarely send a fragment to the wrong tradition, but they often supply the margin that lets it be placed at all. We therefore release the decomposition with every placement and treat one that rests mostly on common features as weak.

5.5 Orphans and the open set

With the calibrated margin, 380 of the 1,766 orphans with five or more legible words are placed (328–408 across the margin's 90% interval), 55% of them with the Kizzuwatna rituals. The test sets show why this needs care. The named-practitioner rituals of CTH 390–469, which no study assigns to Kizzuwatna, are placed mostly with the Kizzuwatna rituals (56 of 79), the only large ritual class: genre before origin. CTH 500 behaves differently: 58 of its 81 placed fragments go to the Hurrian festival groups, in line with HFR's treatment of CTH 500 as festival material (Rieken and Schwemer, 2026).

A closed reference always names a tradition (Koppel et al., 2011). When windows of 10 words from six other genres are scored, the margin rule still places 7%–14% of them, and 41% of windows from festivals of other traditions. Adding the outside genres as classes of their own (a genre gate) recognises 88%–97% of their windows and 62% of the other festivals, at the cost of sending 23% of reference windows away. Applied to the orphans, the gate finds 94 that are closer to an outside genre than to any tradition by more than the margin, 11 of them among the placed. We release them as candidates for re-filing.

The damage-aware text places far fewer orphans than the editor's full text (847 of 2,026 in version A), because many orphans lose most of their words with the restorations. The placements that remain are stable: of the 736 orphans placed in version A that are still scored in C, 681 (93%) keep the same nearest tradition. The loss is in confidence, not in direction.

5.6 Duplicates, not joins: a negative result

A fragment's nearest manuscripts are leads for the philologist, and the question is what kind of lead they are. Against the Konkordanz, the ranking finds duplicates: of 180 recorded duplicate pairs, the partner ranks first in 60 and among the 20 nearest of 16,604 manuscripts in 109 (median rank 8). On the editor's full text the figures look better (median rank 2, 165 of 212 in the top 20), but partly by circularity: editors restore duplicates from each other. Of the 11 suspected joins, only 1 reaches the top 20 (median rank 278). Two pieces of one tablet share their context but, as a rule, not their text, so we tried to learn a ranking for joins (run so far on version A).

TLHdig marks on every line of a joined text the piece it is preserved on. Splitting its joined texts back into their pieces gives 2,335 query pieces from 897 joins, each to be found among 19,821 units. A logistic regression re-ranks the nearest candidates of each piece over twelve pair features: similarity, the similarity of the two units' profiles over traditions and over compositions, word and phrase overlap, the share of shared words spelled alike, the

	top 1	top 5	top 20	median rank
Similarity ranking	11%	30%	48%	24
Join model (re-ranking)	13%	32%	51%	9

Table 5: Finding the other piece of a known join: share of 2,335 pieces whose partner ranks in the top 1, 5 and 20, and its median rank, tested on compositions not used in fitting.

distance of their orthographic rates, the overlap of their preserved sides and columns, the same publication volume, the same excavation series, and size. It is tested on compositions it was not fitted on (five folds grouped by CTH number). The gain is small (Table 5): the partner reaches the top 20 for 51% of pieces, against 48% for similarity alone, although the median rank falls from 24 to 9. On the Konkordanz's suspected joins the model places 4 of 10 in the top 20, against 3.

The weights show why. Similarity (+0.85) and the shared tradition and composition profiles (+0.49, +0.58) favour a join, as do identical spellings (+0.20) and preserved columns (+0.23); word overlap counts against it (−0.26), a weak sign of the expected complementarity. The scribal rates have no measurable effect (−0.07, 95% interval −0.16 to 0.02). The joins already made are those scholars could see, mostly pieces that share text, so they teach a model what duplicates look like. Find spot and date, the evidence one would want, cannot be fitted honestly, because for known joins the Konkordanz often derives them from the join itself. Text alone, we conclude, finds parallels and duplicates but does not tell a join from them; that needs physical evidence.

6 Discussion and conclusion

Genre before origin. A fragment is placed by what it shares with the reference, and the largest shared layer is genre. A ritual from outside every reference tradition will look most like the largest ritual class, as the residue of CTH 390–469 shows, and the open-set test shows the same risk across genres. A placement is therefore "nearest of these", and is only as good as the alternatives in the reference. Each tradition added narrows the gap: with the Luwian rituals in the reference, 10 residue texts go to them rather than to the Kizzuwatna rituals, and the spellings-only run shows that the small traditions gain most when their unanalysed manuscripts are counted.

Archetypes and witnesses. The results suggest a division of labour between two levels of a text, which we call Quantitative Scribal Tradition Mapping (QTM). Borrowing the terms of textual criticism (Maas, 1958; Trovato, 2014), a tradition is an *archetype*: not a reconstructed text, but the vocabulary, formulae and foreign-language recitation that its manuscripts share. Each manuscript, and each orphan, is a *witness* that attests part of it. The archetype is identified by content, above all by whom and what a text names. Spelling marks the witness: kept or collapsed, the variant spellings change nothing at the level of the tradition, in line with the view that Hittite writing habits belong to scribal circles and their schooling (Gordin, 2015). An orphan is a low-coverage witness, and how much of each archetype survives only in orphans is a historical question that unseen-species estimates, as used for lost medieval literature (Kestemont et al., 2022), could address.

Conclusion. Content-based assignment places Hittite ritual and festival fragments in literature-defined traditions well above chance from about ten legible words, recovers most compositions held out whole, and gives a quantitative answer to a contested attribution. The exact decomposition makes each placement checkable: it shows that a decision rests on agents, deities, realia and foreign words rather than on spelling or particles, and flags the short fragments where it does not. The same features find duplicates but not joins (Section 5.6). Code, outputs and the placements with their decompositions are released (anonymised for review); every number in this paper is generated from them.

Limitations

The study depends on one digital edition. TLHdig 0.3 has no word analyses for the Luwian corpus and for many rituals, its morphological analyses are automatic and unvalidated, and its normalised forms are uneven across editors. Words with less than half their signs on the tablet are dropped unless the same text supports them, so short fragments over-represent the formulaic lines that survive, and the half-preserved threshold is a choice we have not varied. Several traditions are small (the smallest has 7 test manuscripts), their recall is uncertain, and held out by composition several are not recovered at all. The traditions themselves are taken from the literature: the method tests them, but cannot find a tradition that no study has proposed. The

idf weights are fitted on the whole reference set. Festival manuscripts also differ by function (outline tablets, day tablets, inventories close to festival text (Cammarosano, 2021)), which word profiles may track alongside tradition. The word categories enter only the analysis of what a decision rests on, not the placements; their rules were checked on a sample, but the meaning classes of nouns rest on a gloss list drafted with a language model that is still to be checked by hand. Finally, every placement is a hypothesis for verification on the tablet; a fragment placed near a known manuscript is a candidate, not a join.

Ethics statement

The corpus is openly licensed. A language model (Claude, Anthropic) was used to write and run the analysis code, to draft the gloss list described in Section 4, to draft the text and to check the references; the authors reviewed all of it and take responsibility for the content. *[Authors to confirm this statement.]*

References

Daliah Bawanypeck. 2005. *Die Rituale der Auguren*. Number 25 in Texte der Hethiter. Winter, Heidelberg.

John Burrows. 2002. 'Delta': A measure of stylistic difference and a guide to likely authorship. *Literary and Linguistic Computing*, 17(3):267–287.

Michele Cammarosano. 2021. *At the Interface of Religion and Administration: The Hittite Cult Inventories*. Number 68 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Birgit Christiansen. 2006. *Die Ritualtradition der Ambazzi: Eine philologische Bearbeitung und entstehungsgeschichtliche Analyse der Ritualtexte CTH 391, CTH 429 und CTH 463*. Number 48 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Maciej Eder. 2015. Does size matter? authorship attribution, small samples, big problem. *Digital Scholarship in the Humanities*, 30(2):167–182.

Shai Gordin. 2015. *Hittite Scribal Circles: Scholarly Tradition and Writing Habits*. Number 59 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Shai Gordin, Gai Gutherz, Ariel Elazary, Avital Romach, Enrique Jiménez, Jonathan Berant, and Yoram Cohen. 2020. Reading Akkadian cuneiform using natural language processing. *PLOS ONE*, 15(10):e0240511.

Shai Gordin, Emma Yavasan, and Avital Romach. 2025. Matters of genre, style and orthography: Hittite scribal habits through the stylometric lens. In Susanne Görke, Adam Kryszeń, Gerfrid G. W. Müller, and Charles W. Steitler, editors, *The Tablets and Their Scribes: The Palaeography of Hittite Cuneiform and Writing Practices in Hittite Anatolia*, number 73 in Studien zu den Boğazköy-Texten, pages 107–123. Harrassowitz, Wiesbaden.

Susanne Görke. 2010. *Das Ritual der Aštu (CTH 490): Rekonstruktion und Tradition eines hurritisch-hethitischen Rituals aus Boğazköy/Ḫattuša*. Number 40 in Culture and History of the Ancient Near East. Brill, Leiden.

Susanne Görke. 2020. Ḫatti, Pala und Luwiya: Ausgewählte palaische und luwische festrituale im vergleich. In Susanne Görke and Charles W. Steitler, editors, *Cult, Temple, Sacred Spaces: Cult Practices and Cult Spaces in Hittite Anatolia and Neighbouring Cultures*, number 66 in Studien zu den Boğazköy-Texten, pages 291–304. Harrassowitz, Wiesbaden.

Susanne Görke. 2022. Hurrian and Luwian elements in the Kizzuwatna religious texts. *Altorientalische Forschungen*, 49:148–157.

Manfred Hutter. 2003. Aspects of Luwian religion. In H. Craig Melchert, editor, *The Luwians*, Handbuch der Orientalistik I/68, pages 211–280. Brill, Leiden.

Tommi Jauhiainen, Heidi Jauhiainen, Tero Alstola, and Krister Lindén. 2019. Language and dialect identification of cuneiform texts. In *Proceedings of the Sixth Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 89–98, Ann Arbor, Michigan. Association for Computational Linguistics.

Patrick Juola. 2008. Authorship attribution. *Foundations and Trends in Information Retrieval*, 1(3):233–334.

Mike Kestemont, Folgert Karsdorp, Elisabeth de Bruijn, Matthew Driscoll, Katarzyna A. Kapitan, Pádraig Ó Macháin, Daniel Sawyer, Remco Sleiderink, and Anne Chao. 2022. Forgotten books: The application of unseen species models to the survival of culture. *Science*, 375(6582):765–769.

Jörg Klinger. 1996. *Untersuchungen zur Rekonstruktion der hattischen Kultschicht*. Number 37 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Moshe Koppel, Jonathan Schler, and Shlomo Argamon. 2011. Authorship attribution in the wild. *Language Resources and Evaluation*, 45(1):83–94.

Silvin Košak. 2026. Konkordanz der hethitischen keilschrifttafeln. hethiter.net/: hetkonk, `https://hethport.net/hetkonk/`. CTH 701 page accessed 2026-10-16.

Silvin Košak, Gerfrid G. W. Müller, Susanne Görke, and Charles W. Steitler. 2026. Catalog der texte der Hethiter (CTH). hethiter.net/: CTH, `https://hethport.net/CTH/`. Accessed 2026-10-16.

Emmanuel Laroche. 1971. *Catalogue des textes hittites*. Klincksieck, Paris.

Paul Maas. 1958. *Textual Criticism*. Clarendon Press, Oxford.

Jared L. Miller. 2004. *Studies in the Origins, Development and Interpretation of the Kizzuwatna Rituals*. Number 46 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Burt L. Monroe, Michael P. Colaresi, and Kevin M. Quinn. 2008. Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, 16(4):372–403.

Gerfrid Müller, Doris Prechel, Elisabeth Rieken, and Daniel Schwemer. 2026. Thesaurus linguarum hethaeorum digitalis (TLHdig) beta version 0.3. Zenodo.

Elisabeth Rieken and Daniel Schwemer. 2026. Corpus der hethitischen festrituale (HFR). Akademie der Wissenschaften und der Literatur Mainz, hethiter.net/: HFR, `https://hethport.net/HFR/`. Accessed 2026-10-17.

Gerard Salton and Christopher Buckley. 1988. Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5):513–523.

Mirjo Salvini and Ilse Wegner. 1986. *Die Rituale des AZU-Priesters*. Corpus der hurritischen Sprachdenkmäler I/2. Multigrafica, Roma.

Walter J. Scheirer, Anderson de Rezende Rocha, Archana Sapkota, and Terrance E. Boult. 2013. Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7):1757–1772.

Fabian Simonjetz, Jussi Laasonen, Yunus Cobanoglu, Alexander Fraser, and Enrique Jiménez. 2024. Reconstruction of cuneiform literary texts as text matching. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13712–13721, Torino, Italia. ELRA and ICCL.

Efstathios Stamatatos. 2009. A survey of modern authorship attribution methods. *Journal of the American Society for Information Science and Technology*, 60(3):538–556.

Rita Strauß. 2006. *Reinigungsrituale aus Kizzuwatna: Ein Beitrag zur Erforschung hethitischer Ritualtradition und Kulturgeschichte*. de Gruyter, Berlin and New York.

Andrea Trameri. 2025. *Kizzuwatna: History of Cilicia in the Middle and Late Bronze Age (ca. 2000–1200 BC)*. Handbuch der Orientalistik I/185. Brill, Leiden.

Paolo Trovato. 2014. *Everything You Always Wanted to Know about Lachmann's Method*. libreriauniversitaria.it, Padova.

Theo van den Hout. 2009. A century of Hittite text dating and the origins of the Hittite cuneiform script. *Incontri Linguistici*, 32:11–35.

Willemijn Waal. 2015. *Hittite Diplomatics: Studies in Ancient Document Format and Record Management*. Number 57 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Mark Weeden. 2011. *Hittite Logograms and Hittite Scholarship*. Number 54 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Ilse Wegner and Mirjo Salvini. 1991. *Die hethitisch-hurritischen Ritualtafeln des (ḥ)išuwa-Festes*. Corpus der hurritischen Sprachdenkmäler I/4. Multigrafica, Roma.

Ilya Yakubovich and Alice Mouton. 2023. *Luwili: Hittite-Luwian Ritual Texts Attributed to Puriyanni, Kuwattalla and Šilalluḥi (CTH 758–763). Volume II: Discussion and Glossary*. Number 72,2 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Emma Yavasan and Shai Gordin. 2025. From clay to code: Transforming Hittite texts for machine learning. In *Proceedings of the Second Workshop on Ancient Language Processing (ALP 2025)*, pages 77–86. Association for Computational Linguistics.

Vasiliy Yugay, Kartik Paliwal, Yunus Cobanoglu, Luis Sáenz, Ekaterine Gogokhia, Shai Gordin, and Enrique Jiménez. 2024. Stylistic classification of cuneiform signs using convolutional neural networks. *it – Information Technology*, 66(1):15–27.